

from a known displacement in local flat-Earth coordinates $(\Delta x_f, \Delta y_f, \Delta z_f)$. Starting at the initial geographic coordinates (Φ_1, Ψ_1, H_1) , the algorithm is used to evaluate the final geographic coordinates (Φ_2, Ψ_2, H_2) and the geodesic bearing change $\Delta\psi_g$ from the **spherical-Earth approximation**,

$$\begin{aligned}
 d &= \sqrt{\Delta x_f^2 + \Delta y_f^2} \\
 \text{if } (d < \varepsilon_t) &\text{ then} \\
 &\quad \Phi_2 = \Phi_1 \\
 &\quad \Psi_2 = \Psi_1 \\
 &\quad \Delta\psi_g = 0 \\
 \text{else} \\
 &\quad \Theta = d / (R_E + H_1 - \Delta z_f / 2) \\
 &\quad \psi_{g1} = \text{atan2}(\Delta y_f, \Delta x_f) \\
 &\quad \hat{x} = \cos \Phi_1 \cos \Theta - \sin \Phi_1 \sin \Theta \cos \psi_{g1} \\
 &\quad \hat{y} = \sin \Theta \sin \psi_{g1} \\
 &\quad \hat{z} = \sin \Phi_1 \cos \Theta + \cos \Phi_1 \sin \Theta \cos \psi_{g1} \\
 &\quad \hat{x}' = -\cos \Phi_1 \sin \Theta - \sin \Phi_1 \cos \Theta \cos \psi_{g1} \\
 &\quad \hat{y}' = \cos \Theta \sin \psi_{g1} \\
 &\quad \hat{z}' = -\sin \Phi_1 \sin \Theta + \cos \Phi_1 \cos \Theta \cos \psi_{g1} \\
 &\quad \hat{r} = \sqrt{\hat{x}^2 + \hat{y}^2} \\
 &\quad \Phi_2 = \text{atan2}(\hat{z}, \hat{r}) \\
 &\quad \Psi_2 = \Psi_1 + \text{atan2}(\hat{y}, \hat{x}) \\
 &\quad C = \hat{x}^2 \hat{z}' \\
 &\quad S = (\hat{x} \hat{y}' - \hat{y} \hat{x}') \cos^2 \Phi_2 \cos^2 (\Psi_2 - \Psi_1) \\
 &\quad \Delta\psi_g = \text{atan2}(S, C) - \psi_{g1} \\
 \text{end if} \\
 H_2 &= H_1 - \Delta z_f
 \end{aligned} \tag{11.12.37}$$

where ε_t is a small tolerance on the order of the square root of machine precision, which is used as a test to avoid the indeterminate arctangent call for vertical flight.

The curve described by Eq. (11.12.37) is a geodesic, which was defined to be a path of zero acceleration. As discussed previously, the geodesics are straight lines within the flat-Earth approximation, but become spatial curves when the formulation is written in geographic coordinates. In the context of the spherical-Earth model, the geodesics are called *great circles*. Remember that the use of Eq. (11.12.37) does not assume that the aircraft follows a geodesic in traveling from point 1 to point 2. The transformation given by Eq. (11.12.37) simply maps the linear geodesics of the local flat-Earth approximation to the great-circle geodesics associated with the spherical-Earth approximation.